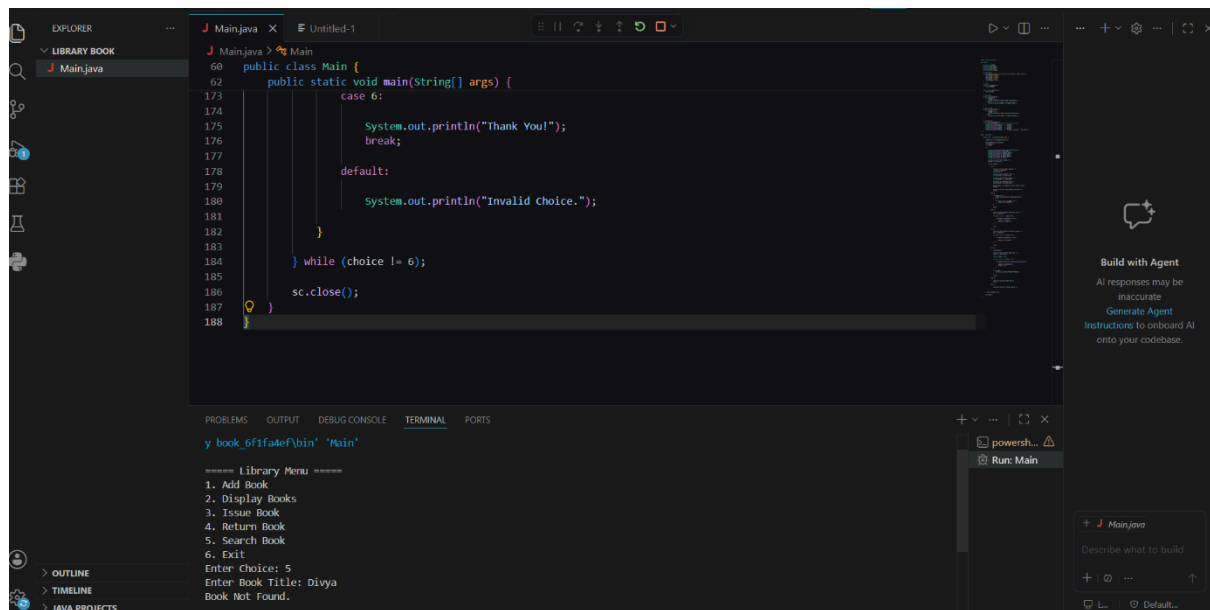


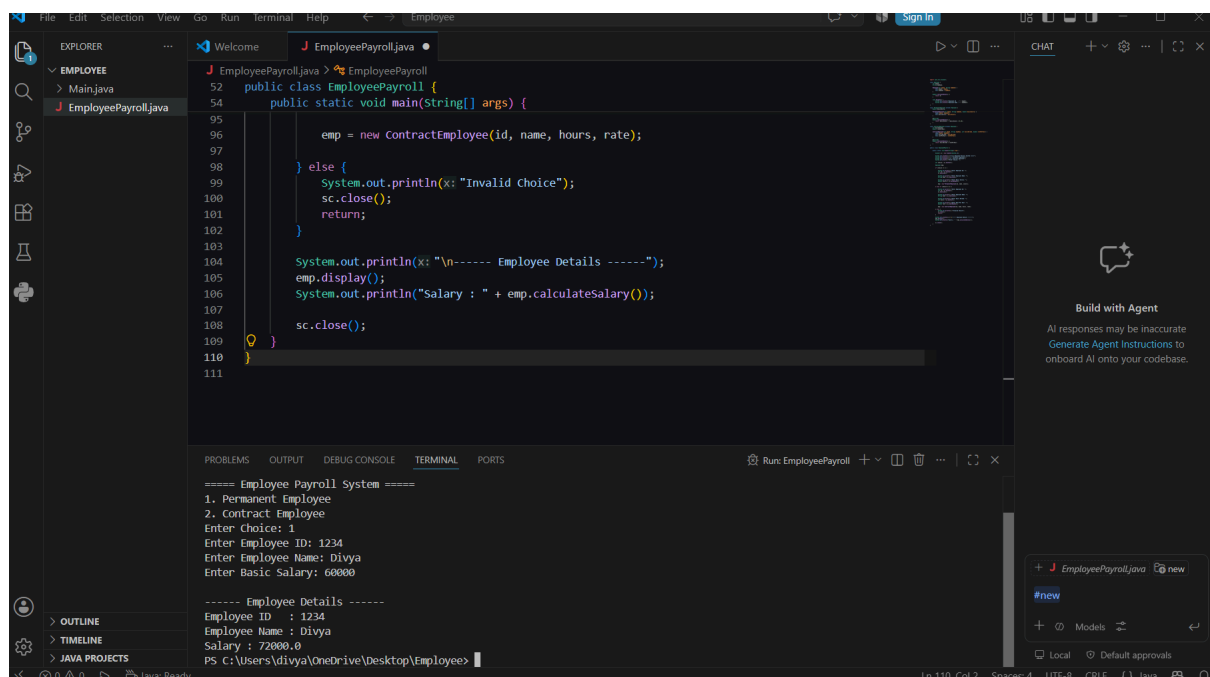
## Question 1: Library Book Management using Classes & Encapsulation



```
60 public class Main {
61     public static void main(String[] args) {
173         case 6:
174             System.out.println("Thank You!");
175             break;
176
177         default:
178             System.out.println("Invalid Choice.");
179
180     }
181 }
182
183 } while (choice != 6);
184
185 sc.close();
186
187 }
```

Library Menu =====  
1. Add Book  
2. Display Books  
3. Issue Book  
4. Return Book  
5. Search Book  
6. Exit  
Enter choice: 5  
Enter Book Title: Divya  
Book Not Found.

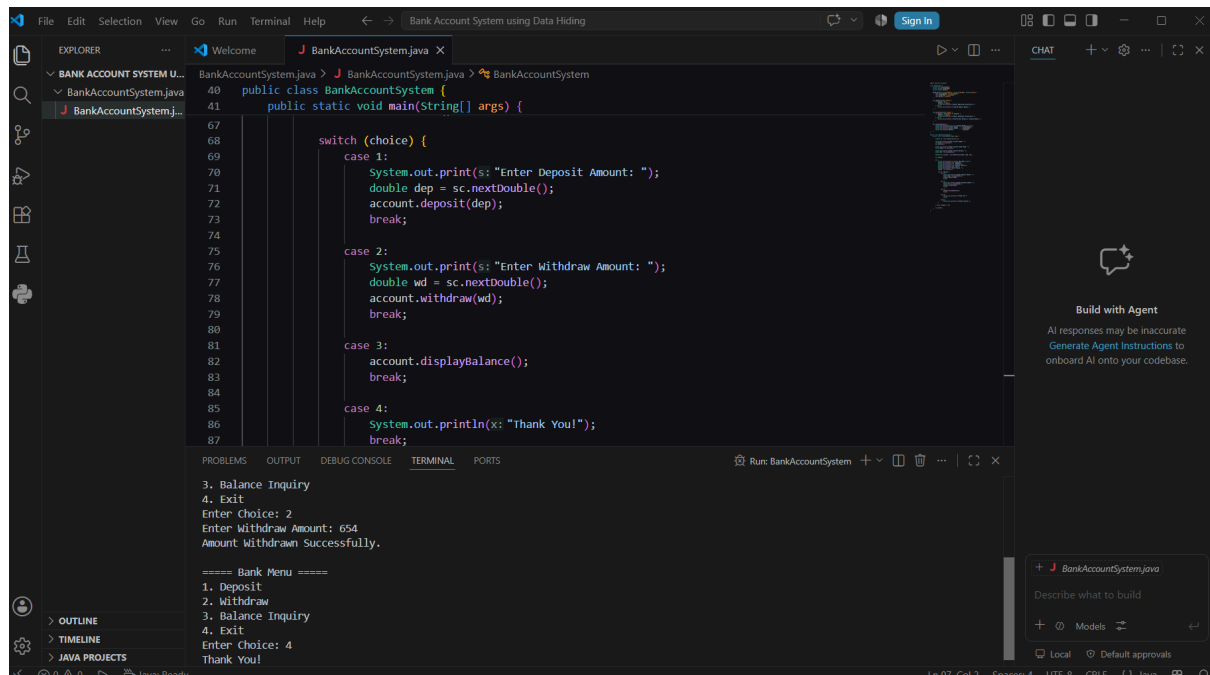
## Question 2: Employee Payroll System using Inheritance



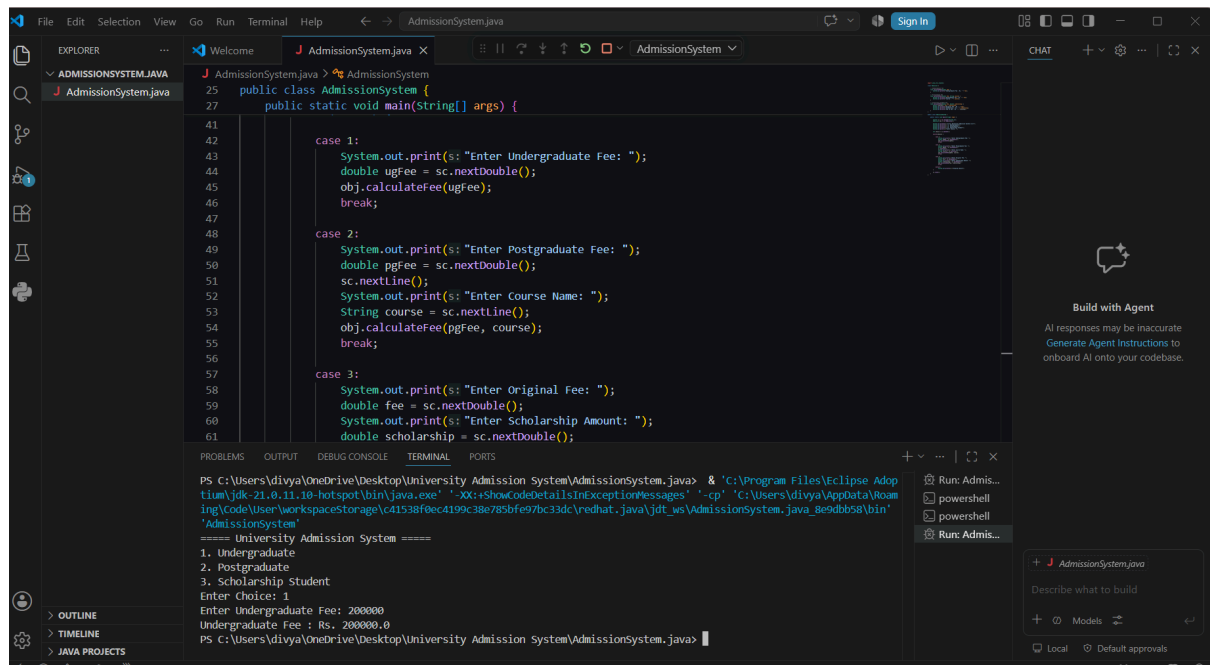
```
52 public class EmployeePayroll {
53     public static void main(String[] args) {
95         emp = new ContractEmployee(id, name, hours, rate);
96
97     } else {
98         System.out.println("Invalid Choice");
99         sc.close();
100         return;
101     }
102
103     System.out.println("\n----- Employee Details -----");
104     emp.display();
105     System.out.println("Salary : " + emp.calculateSalary());
106
107     sc.close();
108 }
109 }
```

Employee Payroll System =====  
1. Permanent Employee  
2. Contract Employee  
Enter Choice: 1  
Enter Employee ID: 1234  
Enter Employee Name: Divya  
Enter Basic Salary: 60000  
----- Employee Details -----  
Employee ID : 1234  
Employee Name : Divya  
Salary : 72000.0

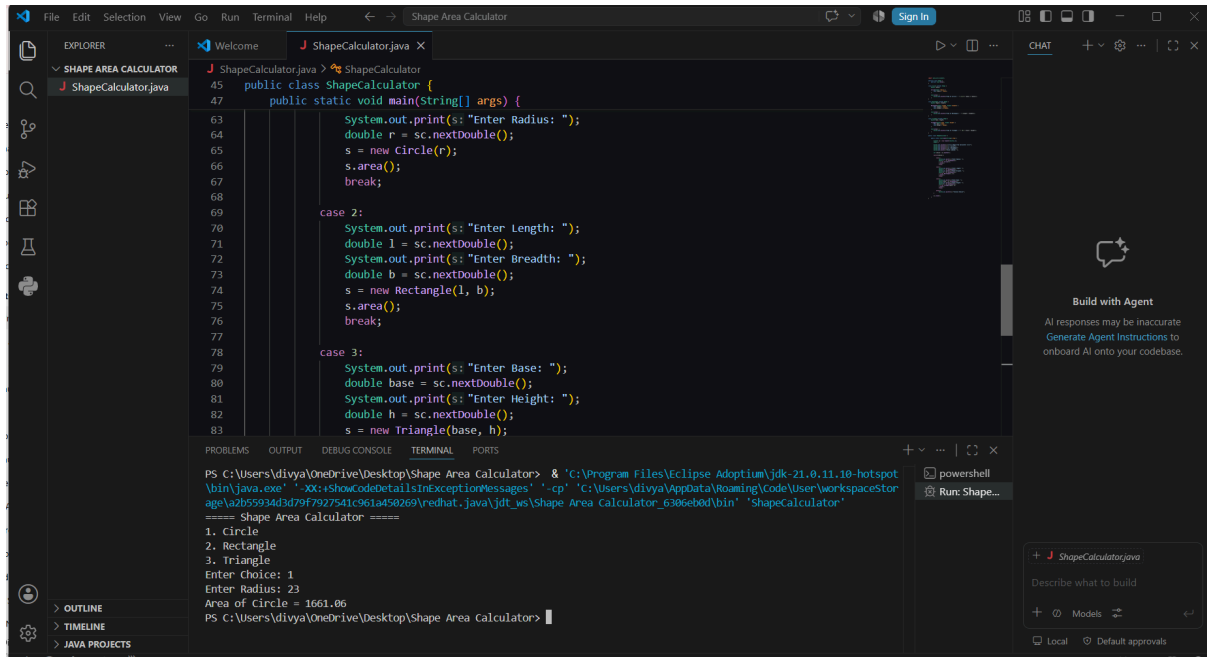
### Question 3: Bank Account System using Data Hiding



#### Question 4: University Admission System using Method Overloading



## Question 5: Shape Area Calculator using Abstract Class & Polymorphism



```
File Edit Selection View Go Run Terminal Help Shape Area Calculator
EXPLORER
  SHAPES AREA CALCULATOR
    ShapeCalculator.java
  J ShapeCalculator.java
  Welcome
  ShapeCalculator.java
    ShapeCalculator
    public class ShapeCalculator {
    45 public static void main(String[] args) {
    47
    63     System.out.print(s: "Enter Radius: ");
    64     double r = sc.nextDouble();
    65     s = new Circle(r);
    66     s.area();
    67     break;
    68
    69     case 2:
    70     System.out.print(s: "Enter Length: ");
    71     double l = sc.nextDouble();
    72     System.out.print(s: "Enter Breadth: ");
    73     double b = sc.nextDouble();
    74     s = new Rectangle(l, b);
    75     s.area();
    76     break;
    77
    78     case 3:
    79     System.out.print(s: "Enter Bases: ");
    80     double base = sc.nextDouble();
    81     System.out.print(s: "Enter Height: ");
    82     double h = sc.nextDouble();
    83     s = new Triangle(base, h);
    84
    85     }
    86     }
    87 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\divya\OneDrive\Desktop\Shape Area Calculator> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.11-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\divya\AppData\Roaming\Code\User\workspacestorage\azb5934d3d79f7927541c961a458269\redhat.java\jdt_ws\Shape Area Calculator_6306eb0d\bin' "ShapeCalculator"
==== Shape Area Calculator ====
1. Circle
2. Rectangle
3. Triangle
Enter Choice: 1
Enter Radius: 23
Area of Circle = 1661.06
PS C:\Users\divya\OneDrive\Desktop\Shape Area Calculator>
```

Build with Agent  
AI responses may be inaccurate.  
Generate Agent Instructions to onboard AI onto your codebase.

+ J ShapeCalculator.java  
Describe what to build.  
+ @ Models  
Local Default approvals